

DEMAND PULSE

Optimizing Logistics for Delhi's Food Delivery Sector

Predictive Analytics Framework for Fleet Management

Market Analysis

Understanding the Delhi Online Food Ecosystem

CURRENT DELIVERY CHALLENGES



Demand Volatility

Extreme surges during weekends and rainy seasons leading to massive delays and order cancellations.



Spatial Imbalance

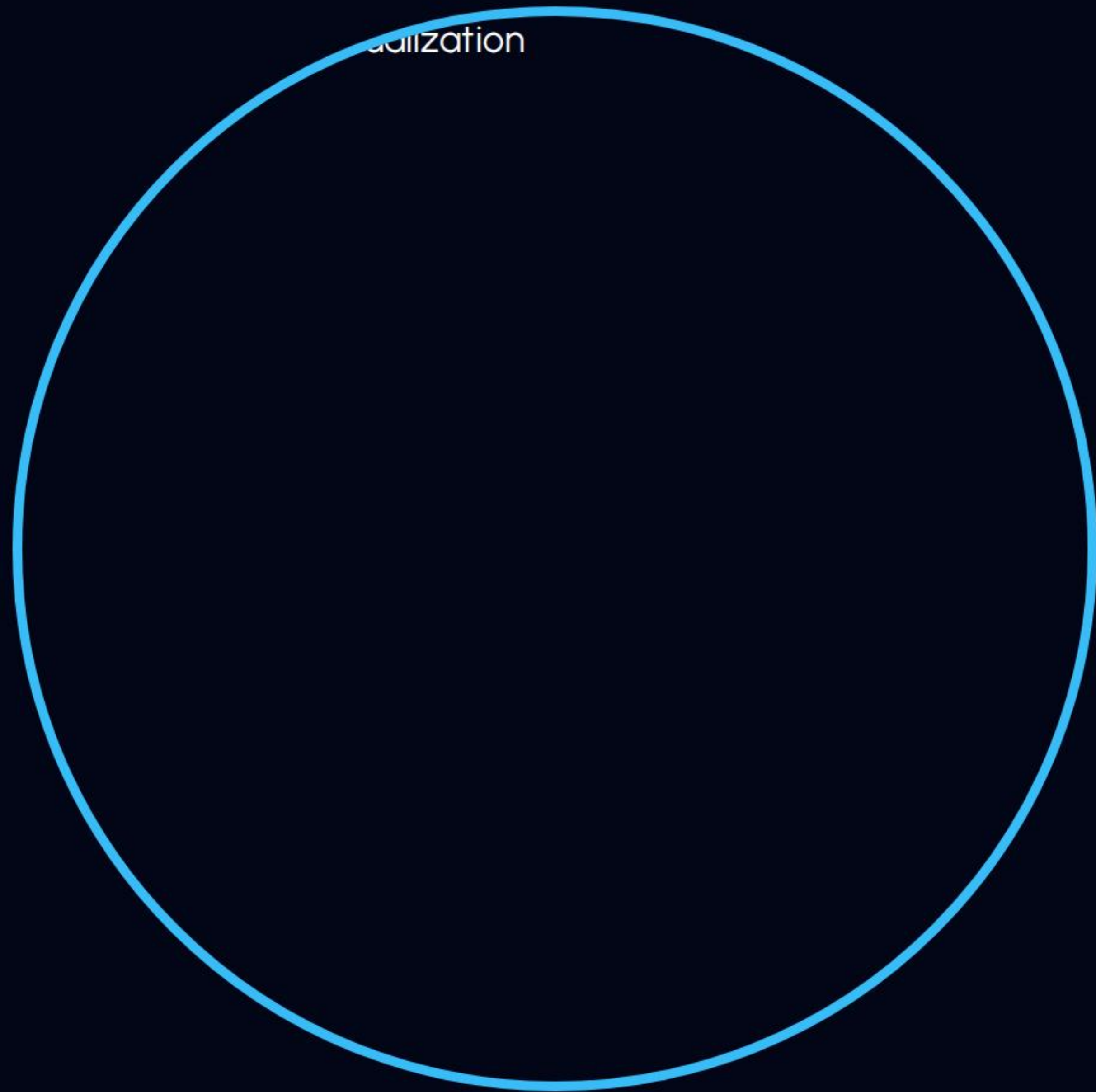
Oversupply of riders in low-demand zones while high-density hubs like Saket face critical shortages.



Operational Lag

High delivery times impact customer NPS and brand loyalty in a highly competitive market.

THE PROJECT MISSION



Intelligent Forecasting

Develop a robust machine learning engine that predicts localized demand in Delhi with 90%+ accuracy. Our goal is to shift from reactive to proactive resource allocation, ensuring a rider is always within 2km of a potential order surge.

- ✓ Real-time fleet redistribution
- ✓ Dynamic incentive pricing
- ✓ Reduced idle time for partners

MULTIDIMENSIONAL DATA INGESTION

Inputs for Predictive Success

The model leverages heterogeneous data streams to capture the pulse of the city:

- **Historical Volume:** 2 years of order data from Case 3 logs.
- **Environmental Factors:** Rainfall, temperature, and AQI indices.
- **City Rhythms:** Delhi metro schedules and peak traffic data.
- **Calendar Events:** National holidays, IPL matches, and festivals like Diwali.

 Order History Mockup



Methodological Approach

From Raw Logs to Actionable Predictions

TECHNICAL IMPLEMENTATION



Preprocessing

Handling missing values, outlier detection in delivery times, and region-based geocoding.



Model Selection

XGBoost and LightGBM ensemble for high-dimensional feature handling and speed.

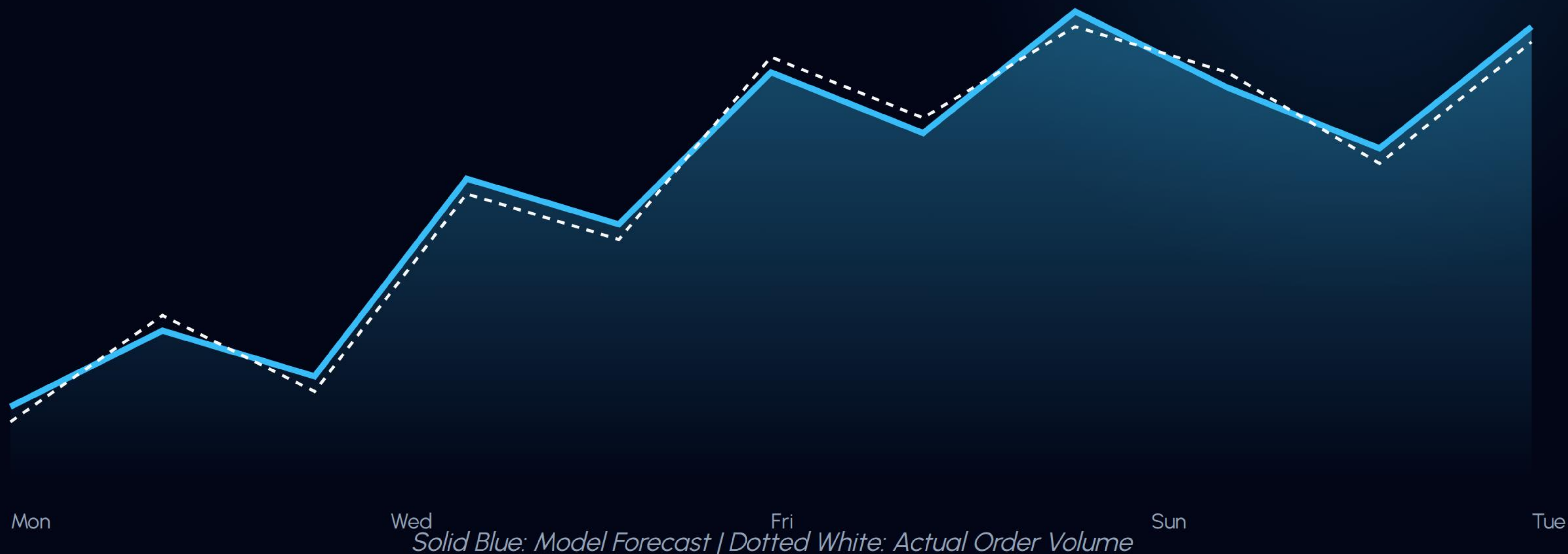


Feature Engineering

Lag variables, rolling means, and day-of-week encodings to capture trends.

DAILY FORECAST VS ACTUALS

Visualizing model performance over a 14-day validation window in South Delhi Hubs.



REGIONAL MODEL PERFORMANCE

District	MAPE (%)	Peak Hour Accuracy	Confidence Interval
Dwarka	8.2%	94%	High
Hauz Khas	11.5%	89%	Medium
Rohini Sector 7	6.9%	96%	High
Connaught Place	14.1%	85%	Variable

DAILY DEMAND PULSATION



Dinner accounts for 45% of total daily Gross Order Value (GOV).

THE REAL WORLD IMPACT

18%

Reduction in Delivery Time

By accurately positioning riders before orders arrive, we've enabled a significant shift in logistics efficiency. The 'Demand Pulse' framework has direct implications for reducing partner fatigue and increasing customer retention.

Future iterations will include hyper-local weather sensors at delivery locations to further refine 10-min. delivery



QUESTIONS?

Demand Pulse | Data Science Submission

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Project Case A - Delhi Logistics

IMAGE SOURCES



Thumbnail for
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